

Asadulla Ravshanbekov

MACHINE LEARNING ENGINEER · LLM SYSTEMS · PRODUCTION AI · BACKEND

📍 Munich, Germany ✉ asadullaravshanbekov04@gmail.com 🌐 github.com/asadiko 🌐 asadulla-ravshanbekov

🇪🇺 Authorized to work in Germany — German national visa, EU Blue Card in progress · Available for on-site, hybrid or remote roles

Summary

Machine learning engineer with 3+ years of experience building production AI systems end to end. Primary developer of an LLM document-intelligence service for automotive and manufacturing enterprise clients — requirement extraction, retrieval-augmented feasibility assessment, and a human-in-the-loop learning system shipped through 219 automated releases. Co-founder and CTO of Olber, a live two-sided marketplace (iOS/Android, 10,000+ registered users, real-money escrow wallet) built and operated in my free time, where I co-led security hardening of the production platform. Strongest where LLM pipelines, async Python backends, and production reliability meet.

Work Experience

Semorai GmbH

Machine Learning Engineer

Heilbronn, Germany | Remote/Hybrid

Jun 2023 – Present

AI document-intelligence platform for automotive and manufacturing enterprise clients; polyglot gRPC microservices with a shared LLM gateway (Azure OpenAI), Qdrant vector search, PostgreSQL and MinIO.

- Built the first generation (v1) of the requirement-extraction service — a 13K-LOC async Python gRPC microservice turning engineering specs into classified, feasibility-assessed requirements — then was primary developer of the v2 rewrite: roughly half of all commits and most PR integration across 219 automated releases (semantic-release, signed Harbor images). Re-architected per-requirement LLM stages into batched pipelines, cutting per-document model calls by roughly an order of magnitude.
- Built the retrieval-augmented feasibility engine that answers “can we build this requirement?” from past-project evidence: concurrent vector retrieval with multi-query expansion, evidence ranking with per-type floors, batched LLM reasoning, and citation-grounded output in which any claim without a source reference is stripped after one corrective retry.
- Designed a two-tier learning loop feeding reviewer corrections back into the pipeline: byte-identical re-uploads resolve by content hash with zero LLM calls; retrieved hints are injected behind a coverage gate. A seven-variant ablation chose the injection format; end-to-end document-classification accuracy went from 6/10 to 9/10 with zero regressions and verified tenant isolation.
- Built the norms/standards extraction subsystem from a root-cause analysis of production output: a prefix-agnostic scanner plus deterministic normalization (canonicalization, range expansion, cross-page merge) across 17 standards bodies, with a cheap LLM filter as the last pass rather than the first — validated on a 71-page OEM document with 212 standard codes.
- Hardened the service for long-running loads: runtime-adjustable admission control with queue heartbeats so multi-minute gRPC streams survive proxies, end-to-end cancellation, resource-aware concurrency limits, and malformed-LLM-output repair at every parse site; progress streaming localized in 6 languages. Contributed 21,000+ lines to a ~1,000-case test suite.
- Co-developed the platform’s central LLM gateway used by every AI service: an OpenAI-compatible gRPC/REST service routing 17 Azure model aliases across load-balanced endpoints, with a declarative provider registry, capability-based parameter normalization (reasoning vs. classic models), an adaptive-timeout circuit breaker, bounded admission control and per-call token/cost attribution.
- Across sibling services: built translation-aware multilingual document comparison (an LLM semantic-equivalence gate that suppresses false diffs between translated documents), and improved RAG-ingestion PDF fidelity with a fuzzy word-anchoring pass that repairs vision-model hallucinations in German compound words.

Olber

CTO & Co-Founder (part-time, outside working hours)

Uzbekistan | Remote

Aug 2025 – Present

Two-sided services marketplace for Uzbekistan, built and operated in my free time alongside my full-time role; live on the App Store and Google Play with 10,000+ registered users. Built by a 3-person founding team.

- Co-built and operate the production platform: React Native/Expo mobile apps, two Fastify/Node microservices, and a self-hosted Supabase/PostgreSQL backend (212 migrations, 115 pgTAP test files) with national-ID (MyID) verification, SMS OTP auth and LLM-based content moderation.
- Co-developed the real-money payment system: card-aggregator integration with HMAC-verified webhooks and an outbox pattern, an append-only escrow ledger with commission holds, daily invariant reconciliation and AES-256-GCM card storage; re-architected the holds model and rolled it to production with no maintenance window.
- Co-led a security overhaul of the live platform: a 37-finding row-level-security audit and penetration testing that uncovered a pre-auth account-takeover chain (plaintext OTPs readable via the public API key), marketplace-wide PII exposure and a signup-verification bypass reproduced safely against production. Remediations shipped as 13 tested migrations, adversarially reviewed to catch three further ship-blockers.
- Co-ran production operations and release: resolved a wallet-ledger drift incident via audited compensating entries to a verified zero-delta invariant; hardened the deploy pipeline (test-gated CI, key-based SSH, backups); publish the apps as App Store seller of record through a code-signed self-hosted OTA channel.

2find

Machine Learning Engineer (part-time)

Riga, Latvia | Hybrid

Nov 2022 – Jun 2023

- Built a food and restaurant recommendation system for an early-stage startup, combining user geolocation, restaurant data and NLP-processed preferences into personalized suggestions; Python, TensorFlow, scikit-learn, NumPy.

1 Stallion Capital

Blockchain Engineering Intern

Remote

May 2023 – Jul 2023

- Developed and deployed Solidity smart contracts for DeFi use cases at a blockchain consulting firm.

Selected Side Projects

All projects below are built in my free time and on weekends, outside of my full-time role.

SSS_AI — retail loss-prevention vision system free-time founder project, in development

Multi-camera shoplifting-detection pipeline: YOLOv8-pose extraction, cross-camera person re-identification (OSNet embeddings, thread-safe identity registry with provisional-ID reconciliation) and a custom three-stream spatio-temporal graph CNN (~1.5M params, confidence-gated adjacency; ROC-AUC 0.96, leakage-aware clip-level k-fold). FastAPI/WebSocket serving on GPU Docker. Companion face-watchlist service with two-tier quality gating and K-of-N temporal confirmation so a single frame can never fire a false alert.

UZBook — transport ticketing infrastructure free-time founder project, in development

B2B reservation/ticketing engine for intercity bus carriers (Amadeus-style inventory): FastAPI/PostgreSQL/Redis with race-free seat holds via atomic Redis Lua scripts, RS256 JWT auth with refresh rotation, HMAC-signed QR tickets with constant-time validation, verified payment webhooks and an immutable audit log — 58 endpoints and six role-based frontend apps on a hardened nginx/Docker deploy.

PadelFriend — court-booking marketplace weekend project, co-developed in a two-person team

Originated the NestJS/Prisma backend (156 endpoints, 38 models: HMAC QR check-in, Elo-style ratings, BullMQ queues) and the Next.js player/owner/admin apps. Built a security-hardened three-step email-verification signup (HMAC-stored codes, timing-safe comparison, enumeration-resistant errors, no user row until completion) with a contract generator emitting typed TypeScript/Swift/Kotlin clients from one source.

Hackathons & Awards

- **Junction 2024, Helsinki — 5th place.** Built the matching engine of a deployed job-matching app: a hand-designed psychometric instrument with multi-trait weighted scoring, cosine-similarity matching against ~50 role profiles, greedy maximum-coverage question selection and multi-board job scraping feeding LLM re-ranking.
- **Junction 2023, Helsinki — 4th place.** Prototyped an LLM website generator: GPT-4 + DALL-E 3 code and asset generation driven by a file-operation protocol, with automated project scaffolding and process-managed hosting.
- **START Hack 2023, St. Gallen — top 10.** NLP moderation system for Supercell's player-safety case: detecting abusive, harassing and self-harm-related language in game chat.
- **HackCodeX / The Raccoons 2023, Riga — top 10.** Real-time sign-language recognition (MediaPipe hand landmarks with a geometric rule classifier covering 24 ASL letters) plus facial-emotion recognition, integrated into a voice-driven assistant.
- **The Raccoons 2022, Riga — Data & Healthcare track winner.** BioCode, a healthcare records and monitoring platform.

Leadership

- **Co-organizer, UzHack 2026 (Riga)** — first international hackathon for the Uzbek student diaspora in the Baltics: 60+ participants, 13 teams, 48 hours. Co-led the full lifecycle over ~3 months, developed institutional partnerships (Uzbek embassy, Vatandoshlar Foundation, IT Park Uzbekistan) and mentored teams.
- **Dean of Students, WAYU (World Association of Youth of Uzbekistan), Riga** (Sep 2025 – May 2026) — advised 200+ students on studies and relocation; built the community's support programs from the ground up.

Education

Riga Technical University

BSc Computer Science

Riga, Latvia

Sep 2022 – Jan 2026

- Thesis: a custom CNN for 7-class skin-lesion classification (including melanoma) on the HAM10000 dermatoscopy dataset — 96.94% test accuracy with 5-fold cross-validation, Optuna hyperparameter optimization and early stopping.

Specialized School No. 50 with Advanced STEM Focus

Advanced Secondary Education with Honors

Tashkent, Uzbekistan

Sep 2011 – May 2022

- Designed and built a working helicopter prototype at age 15, later adopted by the school's engineering department as a cornerstone project for advanced students.
- Advanced specialized curriculum: Calculus, Advanced Algorithms, Physics, Computer Programming, Data Structures.

Technical Skills

- **Languages:** Python (primary), TypeScript, JavaScript, SQL, Solidity; familiar with Swift & Kotlin
- **AI / ML:** LLM application engineering & prompt design (Azure OpenAI, Anthropic Claude, GPT-4/DALL-E), RAG & vector search (Qdrant, pgvector, embeddings, BM25/hybrid), agent & multi-agent systems, LLM evaluation & ablation studies, LangChain, LlamaIndex; deep learning (PyTorch, TensorFlow, scikit-learn); computer vision (YOLO/Ultralytics, pose estimation, ST-GCN, person re-identification, OpenCV, MediaPipe, InsightFace); NLP, recommendation systems, model calibration
- **Backend:** gRPC & Protocol Buffers, REST, WebSocket & SSE streaming, asyncio concurrency, FastAPI, Flask, NestJS, Fastify, Node.js; microservice architecture, message/job queues (BullMQ), webhooks
- **Data & storage:** PostgreSQL, Redis, SQLite, MongoDB, Supabase (Postgres + RLS), Qdrant, MinIO / S3, SQLAlchemy, Prisma, Alembic
- **Frontend & mobile:** React, Next.js, React Native / Expo, Vue, Vite, TypeScript, Tailwind CSS, Zustand, TanStack Query, Redux
- **Infrastructure & DevOps:** Docker (multi-stage, GPU) & Compose, CI/CD (GitHub Actions, GitLab CI, semantic-release), nginx, Linux, Azure, Vercel; testing (pytest, Vitest, pgTAP)
- **Security:** penetration testing & vulnerability assessment, RLS/authorization auditing, auth hardening (JWT/RS256, HMAC, OTP/TOTP, OAuth), AES-256-GCM encryption
- **Spoken languages:** English (fluent), Russian (fluent), Uzbek (native), German (beginner, learning)